Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/20/2007
PI Name:	Narain, Amitabh Ph.D.		
Project Title:	Direct Computational Simulations and Experiments for Internal Condensing flows System-Instabilities/Dynamics in Microgravity and Terrestrial Environments		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
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PI Email:	narain@mtu.edu	**Fax:**	FY 906-487-2822
PI Organization Type:	UNIVERSITY	**Phone:**	906-487-2225
Organization Name:	Michigan Technological University		
PI Address 1:	Mechanical Engineering - Engineering Mechanics		
PI Address 2:	1400 Townsend Drive		
PI Web Page:			
City:	Houghton	**State:**	MI
Zip Code:	49931-1295	**Congressional District:**	1
Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
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Task Description:	Understanding of qualitative phenomena based on experiments supported by accurate computational simulations (steady and unsteady) for quasi-steady annular/stratified internal condensing flows in micro-gravity and terrestrial environments is essential for good design and performance of condensers in applications such as Rankine-cycle based power (nuclear) plants, Thermal Management systems, Looped Heat Pipes, Capillary Pumped Loops, etc. Condenser-sections in these devices often achieve complete or incomplete condensation by the exit under unspecified or specified (as being proposed) exit conditions while the inlet (pressure and flow rate) and wall conditions are typically always known or specified. These condensing flows' attainability/stability of steady flows, intrinsic waviness, noise-sensitivity (primarily due to minuscule transverse vibrations of the condensing surface), and exit-condition sensitivity critically depend on the presence or absence of gravity in driving the condensate. Zero-gravity simulations establish that the parameter zone within which steady flows exist for any space-based condenser, experiencing full or partial condensation with unspecified (free) exit condition (as this is a typical exit condition for condensers in many thermal systems), is very small. Thus achieving steady flows in such space-based condenser employments is very difficult (if not next to impossible) unless new technology – to be suggested by the current and planned research on exit-condition based flow categorizations - is employed. The ground based experiments and supporting simulations (both for terrestrial and microgravity environments) are to explore how exit-condition effects can lead to different steady and unsteady/oscillatory flows. The unsteady/oscillatory flows may arise from changes in flow regime within the condenser or from system-instabilities (due to interaction of condenser with other upstream or downstream components of the system). Although the proposed research involves fundamental investigations of flows inside relatively simpler tube and channel geometries, the qualitative understandings/results and tools that will be obtained are going to be sufficiently fundamental and general in nature that they will provide an important framework for facilitating the design, incorporation, and operation of new realistic space-based or ground-based condensers (that is condensers that employ more complex but realistic geometries such as SWEP/Tranter plate heat-exchangers, shell-and-tube type, etc.) in planned systems. Task Objectives Experimental • Complete experimental investigation of attainability/stability and exit-condition issues for downward and steady internal condensing flows inside a vertical tube. Following this gravity driven condensate flow investigations, initiate similar experimental investigations of gravity-to-shear driven internal condensing flows in inclined-to-horizontal channel (i.e. rectangular cross-section of suitable aspect-ratio) geometries which experience condensation on its bottom wall. • Ensure that the ground based experimental facility is well instrumented and progressively capable of providing more and more detailed information that can be synthesized with results obtained from the computational capability being developed for these flows. • Complete ongoing invention efforts and develop a non-intrusive film thickness sensor (based on principles of fluorescence and fiber-optics) that is capable of measuring time-varying thickness of the dynamic condensate. Verify such an invention by independent thickness measurements. Make progress on employing such sensors for the gravity driven annular flow cases in a vertical tube geometry and/or inclined-to-horizontal channel (i.e. rectangular cross-section of suitable aspect-ratio) geometry that experiences condensation on its bottom wall. Computational • Complete development of accurate simulation tool for laminar vapor/laminar condensate internal condensing flows for channel or vertical tube geometries. • Ensure that the simulation tool accurately locates the interface (under smooth or wavy conditions) for terrestrial and zero-gravity environments and meets various convergence and grid independence criteria for both specified and unspecified exit condition cases. • Ensure that the simulation tool is capable of reliably reproducing known results for classical benchmark problems (such as Nusselt problem of film condensation on a vertical plate, forced vapor flow experiencing film condensation on a horizontal plate, etc.) and, also, capable of providing new non-linear stability results consistent with known experimental results for these classical problems. • The verifications of the two dimensional code on classical problems as well as through successful comparisons of the simulation results for the ongoing ground based experiments are necessary for further enhancements (compressibility effects under specified exit conditions, three dimensional simulations, etc.) of the code. The verifications will also make simulation tool based zero gravity predictions more trustworthy. • Eventually the simulation tool should be reliable enough to help plan proper space based experiments and, also, to help reduce the number of design iterations needed to have a condenser operate successfully in a zero gravity environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reliable existing zero-gravity simulations indicate that the parameter zone within which steady flows exist for any space-based condenser that experiences full or partial condensation with an unspecified (free) exit condition (as this is a typical exit condition for condensers in many thermal systems) is often non-existent to small. Thus achieving steady flows in any space-based condenser employments is very difficult (if not next to impossible) unless new technology – to be suggested by the current and planned research on specified exit-condition scenarios - is employed. Earth Benefits The exit condition based flow categorizations and experimental results are also going to be useful for many new terrestrial systems (such as two-phase electronic or chip cooling) that need to employ a condenser with a steady and stable flow for the thermal system. While it is true that many terrestrial systems can be made to function by sufficient number of trial and error experiments, the ground based results from this research will help significantly in innovation and improvement of superior new technologies for such systems. For example, a specified exit condition mode of

	operation is expected to allow steady and stable operation of many terrestrial condensers in novel systems (electronic cooling, etc.) while concurrently cutting down on the system's start up times. Also, the proposed development of film thickness sensor will be valuable in the measurement processes that are important to the development of many terrestrial applications that involve film cooling, film evaporation, film flows in chemical process industries, etc.
Task Progress:	Reported experimental and computational results confirm that both the flow features and heat transfer rates inside a condenser depend on the specification of inlet, wall, and exit conditions. When exit pressure is changed from one steady value to another, the changes required of the interior vapor flow towards achieving a new steady duct flow are such that they do not demand removal of the new exit pressure imposition back to the original steady value. This behaviour of condensing flow is unlike that of any incompressible single phase duct flow which cannot allow imposition of an exit pressure other than the "required" value. Instead, new steady flows may be achieved for different steady exit pressure prescriptions (termed category I flows) through appropriate changes in the vapor/liquid interfacial configurations and associated changes in interfacial mass, heat transfer rates (both local and overall), and other flow variables. This special feature of these flows is for the commonly occurring large heat sink situations for which the condensing surface temperature (not heat flux) remains approximately the same for any given set of inlet conditions while exit condition changes. In this report's experimental context of flows of a pure vapor that experiences film condensation on the inside walls of a vertical tube, the reported results provide important quantitative and qualitative understanding. The theorotical and experimental results presented in this report allow us to propose important exit-condition based categorization of these flows. Of these, category II flows, which are defined to be the cases for which exit pressures are left unspecified, have typically been assumed to hold for most actual steady realizations of gravity driven as well as zero-gravity flows. However it is shown here that steady flows under self selected exit pressure conditions are achieved under these conditions mostly for gravity driven flows and, even then, only if special hardware arrangements are made that allow vapor to freely choose an exit pressure. If this is not so, one often has an inadvertent category I flow without the explicit knowledge of the imposed exit pressure condition. It is theoretically shown here that though this category II steady flow situation in the absence of specification of exit condition constitutes an "ill posed" steady boundary value problem, under special arrangements, steady flows may still be experimentally realized. This fact is also validated by unsteady theory which shows that solutions from many different initial conditions may tend to a unique steady attracting solution (as time tends to infinity) for some of the internal and/or external flows. These "attractors" exist and are strong for gravity driven flows and are often weak to non-existent for zero gravity or horizontal shear/pressure driven flows (both for internal and external flows). In microgravity, the "attractors" for category II internal flows are weak in the sense that they significantly lose their strength of "attraction" after a certain downstream distance – after which the flows are indeterminate if exit pressure remains unspecified. For these microgravity situations, the remedy is to run condensers under suitably specified inlet and exit pressures (category I conditions) as well as a proper cooling strategy (i.e. proper wall temperature variations). The details of our recommendations on how to run space based condensers under category I flow conditions is qualitatively described in the report but actual experimental/theoretical details are part of ongoing research and future reports.
Bibliography Type:	Description: (Last Updated: 06/28/2013)
Articles in Peer-reviewed Journals	Narain A, Kurita JH, Kivisalu M, Kulkarni SD, Siemionko A, Ng TW, Kim N, Phan L. "Internal condensing flows inside a vertical pipe – experimental/computational investigations of the effects of specified and unspecified (free) conditions at exit." Journal of Heat Transfer, 2007 Oct;129(10):1352-72. http://dx.doi.org/10.1115/1.2755063 , Oct-2007
Articles in Peer-reviewed Journals	Phan L, Narain A. "Non-linear stability of the classical Nusselt problem of film condensation and wave effects." Journal of Applied Mechanics, 2007 Mar;74(2): 279-90. http://dx.doi.org/10.1115/1.2198249 , Mar-2007
Articles in Peer-reviewed Journals	Ng TW, Narain A. "Fluorescence and Fiber-Optics Based Real-Time Thickness Sensor for Dynamic Liquid Films." Submitted (after good IMECE07 review) for publication in ASME Journal of Heat Transfer, 2007. expected publication June 2008. , Jun-2008
Articles in Peer-reviewed Journals	Kulkarni SD, Narain A, Mitra S, Phan L. "Forced Flow of Vapor Condensing over a Horizontal Plate (Problem of Cess and Koh*) Part I – Steady Solutions of the Full 2D Governing Equations." Submitted (after good IMECE07 reviews) for publication in ASME Journal of Heat Transfer, expected publication June 2008. , Jun-2008
Articles in Peer-reviewed Journals	Kulkarni SD, Narain A, Mitra S. "Forced Flow of Vapor Condensing over a Horizontal Plate (Problem of Cess and Koh*) Part II – Unsteady solutions of the full 2D governing equations yielding steady "Attractors" and their unsteady responses to initial disturbances and noise." Submitted (after very good IMECE07 reviews) for publication in ASME Journal of Heat Transfer, expected publication June 2008. , Jun-2008
Articles in Peer-reviewed Journals	Narain A, Phan LA, Wang X, Kurita JH, Siemionko A, Ng TW, Kulkarni SD. "Direct computational simulations and experiments for film condensation inside tubes and channels." Ann N Y Acad Sci. 2006 Sep;1077:471-507. PMID: 17124142 , Sep-2006
Papers from Meeting Proceedings	Narain A, Kivisalu M, Siemionko A, Kulkarni SD. "Internal Condensing Flows inside a Vertical Pipe – Experimental/Computational Investigations of the Effects of Specified and Unspecified (Free) Conditions at Exit." Presented at IMECE07 Symposium on Gas Liquid and Phase Change Flows, Seattle, Washington, Nov. 11-16, 2007. Paper IMECE2007-41306, published in IMECE07 Proceedings (CD), November 2007. , Nov-2007
Papers from Meeting Proceedings	Ng TW, Narain A. "Fluorescence and Fiber-Optics Based Real-Time Thickness Sensor for Dynamic Liquid Films." Presented at IMECE07 Symposium on Gas Liquid and Phase Change Flows, Seattle, Washington, Nov. 11-16, 2007. Paper IMECE2007-41304, published in IMECE07 Proceedings (CD) for Symposium on Gas Liquid and Phase Change Flows, Seattle, Washington, November 2007. , Nov-2007

Papers from Meeting Proceedings	Kulkarni SD, Narain A, Mitra S. "Steady Attractors for Forced Flow of Vapor Condensing over a Horizontal Plate (Problem of Cess and Koh)* and their Unsteady Responses to Initial Disturbances and Noise." Presented at IMECE07 Symposium on Gas Liquid and Phase Change Flows, Seattle, Washington, Nov. 11-16, 2007. Paper IMECE2007-41311, published in IMECE07 Proceedings (CD) for Symposium on Gas Liquid and Phase Change Flows, November, 2007. , Nov-2007